

Matrix Theory: Lecture 6

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2025-10-01



Outline

- Repetition: Jordan Normal form
- Cayley-Hamilton's Theorem
- The minimal polynomial
- Multiplicity and diagonalization



The General Jordan Normal Form

Def. For any $\lambda \in \mathbb{K}$ the **Jordan block** $J_k(\lambda)$ of size k looks like:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

thus λ on the main diagonal and $k - 1$ ones on the next. The rest is zero.
For size 1 we have only λ :

$$J_1(\lambda) = \begin{pmatrix} \lambda \end{pmatrix}.$$



The General Jordan Normal Form

Theorem 7.8

For every operator $\mathcal{A}$ there exists a basis consisting of chains of form

$$\begin{array}{ccccccc} (\mathcal{A} - \lambda_1 I)^{k_1-1} v_1, & (\mathcal{A} - \lambda_1 I)^{k_1-2} v_1, & \cdots, & (\mathcal{A} - \lambda_1 I) v_1, & v_1 \\ (\mathcal{A} - \lambda_2 I)^{k_2-1} v_2, & (\mathcal{A} - \lambda_2 I)^{k_2-2} v_2, & \cdots, & (\mathcal{A} - \lambda_2 I) v_2, & v_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{array}$$

with the conditions $(\mathcal{A} - \lambda_i I)^{k_i} v_i = 0$. All λ_i are eigenvalues and some of them may be equal. Relative to this basis the operator has matrix

$$J = \begin{pmatrix} J_{k_1}(\lambda_1) & O & \cdots & O \\ O & J_{k_2}(\lambda_2) & \cdots & O \\ \cdots & \cdots & \cdots & \cdots \\ O & O & \cdots & J_{k_m}(\lambda_m) \end{pmatrix},$$

which is called the **Jordan form** of the operator.

The Characteristic Polynomial

Theorem 7.12

If $\lambda_1, \dots, \lambda_n$ are all roots of $p_A(\lambda)$ (calculated with multiplicity), that is

$$p_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

then $\det A = \lambda_1 \lambda_2 \dots \lambda_n$ and $\operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$. In particular for $n = 2$ we have

$$p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$$



The Cayley-Hamilton Theorem

Theorem 8.1

If we multiply n (upper) triangular matrices $A^{(i)}$ of size n for which $A_{ii}^{(i)} = 0$ for $1 \leq i \leq n$ then their product is equal to zero:
 $A^{(1)} \dots A^{(n)} = O.$

$$\begin{pmatrix} 0 & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & * \end{pmatrix} \begin{pmatrix} * & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & * \end{pmatrix} \cdots \begin{pmatrix} * & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} = O.$$



The Cayley-Hamilton Theorem

Theorem 8.2 (Cayley-Hamilton)

Let A be a square matrix and $p_A(\lambda) = |\lambda I - A|$ its characteristic polynomial. Then

$$p_A(A) = O.$$

An incorrect argument: $p_A(A) = |AI - A| = 0$

Why is this incorrect?

$p_A(\lambda) = |\lambda I - A| = \lambda^n + a_{n-1}\lambda^{n-1} + \dots a_1\lambda + a_0$ takes a **number** λ and gives a new **number** $p_A(\lambda)$.

$p_A(X) = X^n + a_{n-1}X^{n-1} + \dots a_1X + a_0I$ takes a **matrix** X and gives a new **matrix** $p_A(X)$.

$f(X) = |XI - A|$ takes a **matrix** X and gives a **number** $f(X)$.



The Minimal Polynomial

Def. A polynomial

$$p(x) = x^k + a_{k-1}x^{k-1} + \dots + a_0$$

of degree $k \geq 1$ is called the **minimal polynomial** of A if $p(A) = O$ and $r(A) \neq 0$ for any non-zero polynomial $r(x)$ of degree less than k . We denote it $\pi_A(x)$. In other words,

$$\pi_A(x) = x^k + a_{k-1}x^{k-1} + \dots + a_0$$

$$\pi_A(A) = A^k + a_{k-1}A^{k-1} + \dots + a_0I = O$$

and k is as small as possible.



The Minimal Polynomial

Theorem 8.3

If $p(A) = O$ then $\pi_A(x)$ is a divisor of $p(x)$. In particular the minimal polynomial is uniquely determined.

Corollary 8.4

$$\pi_A(x) = (x - \lambda_1)^{s_1} (x - \lambda_2)^{s_2} \cdots (x - \lambda_l)^{s_l},$$

where λ_i are different eigenvalues and $s_i \leq n_i$, where n_i is the multiplicity of λ_i in the characteristic polynomial.



The Minimal Polynomial

Theorem 8.5

For every square matrix A

$$\pi_A(x) = (x - \lambda_1)^{s_1} (x - \lambda_2)^{s_2} \cdots (x - \lambda_l)^{s_l},$$

where l is the number of different eigenvalues of A ,
and for every such eigenvalue λ_i the number s_i is equal to the size of the largest Jordan block containing λ_i on the diagonal in its Jordan form J .



The Minimal Polynomial

Suppose

$$J = \begin{pmatrix} J_{k_1}(\gamma_1) & 0 & \dots & 0 \\ 0 & J_{k_2}(\gamma_2) & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & J_{k_m}(\gamma_m) \end{pmatrix}$$

Since J is block diagonal

$$\pi_A(J) = \begin{pmatrix} \pi_A(J_{k_1}(\gamma_1)) & 0 & \dots & 0 \\ 0 & \pi_A(J_{k_2}(\gamma_2)) & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \pi_A(J_{k_m}(\gamma_m)) \end{pmatrix}$$

Then

$$\pi_A(J_{k_i}(\gamma_i)) = (J_{k_i}(\gamma_i) - \lambda_1 I)^{s_1} \cdots (J_{k_i}(\gamma_i) - \lambda_l I)^{s_l},$$

and if $\gamma_i = \lambda_j$

$$(J_{k_i}(\gamma_i) - \lambda_j I)^{s_j} = J_{k_i}(0)^{s_j}$$



Multiplicity

Theorem 8.6

The following conditions on a matrix A are equivalent:

- 1 The matrix is diagonalizable.
- 2 Its Jordan form contains only blocks of size 1.
- 3 The minimal polynomial has no multiple roots.
- 5 $p(A) = O$ for some polynomial $p(x)$ which has no multiple zeros.

